



CERTIFICATE OF TESTS

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Printed: 05/16/2024

Heat: 2136634
Material: 5L-X52 4-1/2X.337 (15.00#)BR,DRL,PEB (11000102)
Vendor: Nucor Hickman - Armorel AR US (180436)

Purchase Order: 26632645
Sales Order: 1001000690
MFG Date: 02/28/2024

CHEMICAL PROPERTIES

	C	MN	P	S	NI	CR	CU	MO	SI	TI	V	CE
Ladle	0.0400	1.0800	0.0130	0.0013	0.0400	0.0500	0.1300	0.0100	0.2100	0.0130	0.0040	0.1118
	Al: 0.0300		Sn: 0.0050		N: 0.0060		B: 0.0000		Ca: 0.0020		Nb: 0.0320	
Chem	0.0597	1.1965	0.0112	0.0000	0.0381	0.0589	0.1289	0.0095	0.2220	0.0125	0.0045	0.1392
	Al: 0.0333		Sn: 0.0048		N: 0.0113		B: 0.0002		Ca: 0.0020		Nb: 0.0375	
Chem	0.0547	1.1858	0.0115	0.0002	0.0382	0.0586	0.1299	0.0096	0.2185	0.0123	0.0042	0.1336
	Al: 0.0322		Sn: 0.0048		N: 0.0124		B: 0.0003		Ca: 0.0020		Nb: 0.0368	

Acceptance Criteria: CEPCM = 0.2500

MECHANICAL PROPERTIES

YS (ksi): 67.40 TS (ksi): 72.70 EL (%) : 31.00 HRB: 86.00 Y/T(%): 0.93

CHARPY TEST RESULTS (BODY)

Orientation : TRANSVERSE	Specimen(S)	Energy (ft-lb)	%Shear
Spec. Size (in) : .394"X.197"X2.165"	1	79.00	100.00
Test Temp (°F) : -20	2	75.00	100.00
Location : 3T90	3	76.00	100.00
	Average	76.67	100.00

Acceptance Criteria Min. (ft-lb) : 10

COMMENTS

MINIMUM HYDROSTATIC TEST PRESSURE (PSI) : 4,700 PER 5 SECONDS

MANUFACTURED TO THE 46TH EDITION OF API 5L SPECIFICATION
MANUFACTURED IN THE USA-HIGH FREQUENCY WELDED PRODUCT
WELD SEAM ANNEALING AT 1600 DEGREES F.MINIMUM
ULTRASONIC TESTED TO N-10 NOTCH 1/8 DRILLED HOLE
FLATTENING TEST PASSED
1 INCH L90 STRIP SPECIMEN
THIS MATERIAL MEETS :ASTM A53B-22,ASME SA53B-21 LATEST EDITION
API 5L PSL 2 GRADE X52M
MEETS HARDNESS REQUIREMENTS OF NACE MRO 175/ISO 15156-2, CLAUSE A.2.1.1
MEETS ISO 10474 3.1.B/EN 10204: 3.1 / STANDARDS ARE LATEST REVISION
MELTED AND MANUFACTURED IN THE U.S.A.
STEEL SUPPLIER: Nucor Hickman - Armorel AR US
COIL MANUFACTURER: Nucor Hickman - Armorel AR US

The above analyses and tensile properties are correct to the best of my knowledge and belief.


Quality Control Dept